Quotation No : Q23000125

Date : 03-Apr-2023

Manpasand Namkeen
D-29 & D-30 A, Panki Industrial Area
KANPUR
UTTAR PRADESH

Kind Attention : Mr./Ms. Shubham

Dear Sir,

We understand that in industry, reliability is not just a requirement,it is the foundation of success.
That is why Absstem Industrial PSA Generator is more than just a product; it is a commitment to excellence, efficiency, and uninterrupted performance.

Every component is designed with precision and innovation, ensuring that each molecule of oxygen and nitrogen is delivered with the utmost purity.

With fully automated operation, our system seamlessly integrates into your processes, reducing manual intervention and maximizing productivity.

Moreover, our industry-leading air ratio ensures that you achieve the highest efficiency, optimizing energy consumption while maintaining peak performance.
And because we believe in standing by our craftsmanship, we offer an unmatched-10 year warranty on sieves, giving you the peace of mind that your investment is protected for the long run.

We have attached the detailed quotation for your review.

Should you have any questions or require further customization to meet your specific needs, please do not hesitate to reach out.

We are here to provide not just a product but a solution that breathes life into your operations.

We look forward to the opportunity to serve you and build a lasting partnership.

Thanks and Regards,

For Absstem Industrial Solutions Pvt. Ltd.

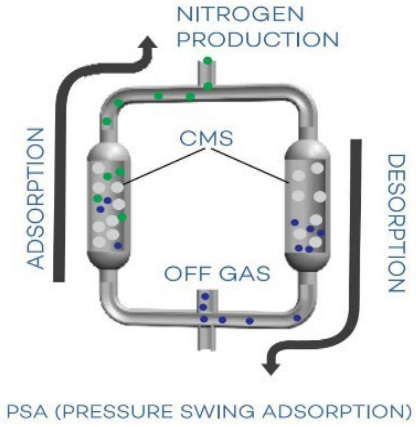
Sushil Kumar
Senior Sales Engineer
9289722208
www.absstem.com

nGEN POWERED BY ABSSTEM INDUSTRIAL

Industrial PSA Nitrogen Generator



PSA Technology



PSA (PRESSURE SWING ADSORPTION)

Industrial Nitrogen Generator mainly consists of two adsorbers filled with carbon molecular sieves. Pressurized air is passed through one of these adsorbers, carbon having the property to attract oxygen, adsorbs oxygen from air and Nitrogen is allowed to pass through and move into the storage tank as the final product. As the adsorber approaches saturation, the air is allowed to pass through the other adsorber while this adsorber is being regenerated to ensure a continuous supply of Nitrogen. This changeover takes place through a specially designed microcontroller to ensure a hassle-free operation. The quality of Nitrogen produced is monitored by a Nitrogen analyser.

Key Features

- Shipped ready for installation
- 5 minutes to run and never have to stop it!
- Easy installation
- Low operating expenses compared to gas cylinders/liquid Nitrogen
- Maintenance costs are very less
- Embedded Safety Interlocks
- Space-saving design
- Eliminates all hazards which are there with gas cylinder or liquid Nitrogen
- Longevity: at least 10 years

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Plot No. 591, Sector 8, Phase IV
IMT Manesar, Gurugram, Haryana - 122051
Contact - 1800 3010 3394 Email - indsales@absstem.com
CIN: U29256HR2022PTC103387

Description

a.	Product Name	Nitrogen Gas Generator
b.	Model Range	**NGEN**
	Technology	PSA (Pressure Swing Adsorption)

Specifications

a.	Nitrogen Flow Rate	As per Performance Sheet
b.	Nitrogen Purity	As per Performance Sheet
c.	Nitrogen Delivery Pressure	5.5 Barg (Higher Pressure Options Available)
d.	Ambient conditions	
	Temperature	0 to 50 Deg. Celsius
	Relative Humidity	up to 80 % (Max.)
	Altitude	up to 1000 m above Sea level

Scheme Offered – (PSA Only)

This scheme will comprise of

- Air package with inlet air filtration system
  (Air compressor, Air dryer, Water Separator, Oil filters)
- Industrial Nitrogen generator
- Storage Tank
- Nitrogen Analyzer
- Alarm panel

Unique Features of Absstem

a. All the fabricated items are cleaned and degreased following our strict quality policy.
b. All the Instruments are specially manufactured to make them suitable for Nitrogen gas.
c. Guaranteed 10+ years of healthy life.
d. We provide the purest **Nitrogen** in Industry.
e. Data logging and Graphical Reports for all parameters available.
f. Test certificates of all instruments are available on demand.

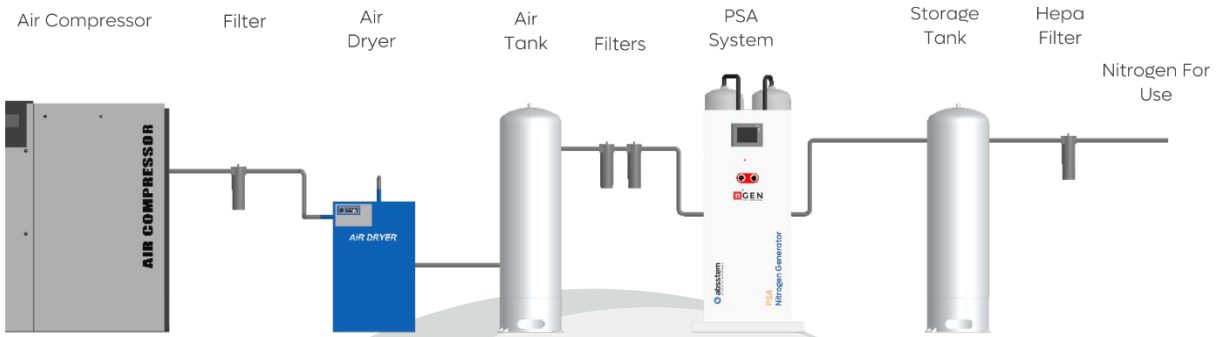
Onsite Gas Generation vs Liquid Tank/Cylinder Alternative

Non-stop availability: 24 hours a day, 7 days a week
Easy integration within existing installations
Your own independent supply of industrial gas
Significant economies for scale and lower operational costs: no rental charges, logistical expenses and bulk user evaporation losses
No safety hazards when handling high-pressure cylinders

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Equipment Description

The Industrial Nitrogen Generator consists of



Air Compressor	An Air compressor is used to compress the atmospheric air to a pressure required for the separation of **Nitrogen** from the air.
Refrigeration dryer	Compressed air then passes through a refrigerated dryer to remove moisture (which is harmful to sieves) from the air.
Air Tank	Dry air is collected in an air tank to remove the pulsations which serve the dynamic (changing) requirement of air by the PSA system.
Multistage Filtration system	Before air enters the PSA system, it is passed through multistage filtration removing moisture, particulates and oil content to the specified levels.
PSA system	Pressurized air is passed through PSA Towers filled with Molecular sieves to separate **Nitrogen** from Air.
Storage Tank	Industrial Grade **Nitrogen** produced from NGEN is stored in this tank to supply a continuous flow of **Nitrogen**.
Nitrogen Analyzer	Nitrogen produced is monitored by a **Nitrogen** analyser. In case, O2 purity goes below the specified level, it gives an alarm and switches it to vent mode.

HD Touch Screen Display

(User-friendly touchscreen interface for easy operation and maintenance)

Absstem industrial Nitrogen generator complies with all industry standards and regulation

- Alarm management system
- Plug and play
- Fully automatic
- Process data display
- Consumables running hours display
- Data logging via USB drive
- Graphical representation



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Additional upgrades

- Digital flow meter with totaliser
- High-pressure Nitrogen tank
- Cylinder manifold and alarm management system
- Remote monitoring

Plant Operation

- Unit operates 24-hrs a day, 7-days a week reliably, automatically & safely

Automation

- The Plant is fully automatic and needs only electrical power for running the Air compressor
- The machine automatically enters into standby mode when the level in the Storage Tank reaches its set limit. This helps to increase the life of the Air compressor.

International Quality standards

A strict quality procedure, following the leading international pharmacopoeia and Industrial directives, is followed in the selection of all mechanical and electrical instruments, vessel fabrication and designing of Industrial Nitrogen generators.

Itemised Supply Description

No.	Item description	Make	Qty
a	Air Compressor with Filters	ABSSTEM APPROVED	Quoted Separately
b	Refrigeration Dryer	ABSSTEM APPROVED	Quoted Separately
c	Air Tank	ABSSTEM	Quoted Separately
d	Set of Filters	ABSSTEM	1 no.
e	PSA unit (NGEN)	ABSSTEM	1 no.
f	Electrical Power Panel with PLC and all cabling	ABSSTEM & MITSUBISHI	1 no.
f	Digital Nitrogen Analyzer	ABSSTEM	1 no.
g	Storage Tank	ABSSTEM	Quoted Separately
h	Outlet product gas Filter	ABSSTEM	1 no.

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Electrical Protection

The client should make proper arrangements for earthing/grounding systems for the Industrial **Nitrogen** Generator at their facility.

Enclosure Panel

- Ergonomically designed metal enclosure with powder-coated finish for longer life.
- High-strength IP-rated panel fittings from Germany for a robust enclosure.
- Fan filters from **Rittal** are IP-55 rated making the electrical enclosure's internal environment dust-free.
- Easy connectivity to any place of installation with req. port connections.
- High-strength base frame designed for higher load and easy transportation.
- Vessels are designed per ASME section VIII DIV I. Vessels are epoxy paint coated which is considered good for longevity and corrosion resistance.
- All electrical items are CE certified with operating voltage at 24 VDC

Annual Maintenance Contract (AMC)

Our in-house support Absstem shield is 24*7 open and we assure a timely service maintenance schedule. Per the schedule mentioned in the operational manual, we suggest the operator change the filters by following a simple protocol.

It is very easy to change or replace as our commissioning engineer plus our commissioning engineer and tech support office will always guide you about the activity.

Best ROI Guaranteed

ROI within 6 to 18 months when compared to conventional methods

Generator performance depends on the component selection. Our processes make this job easier as all equipment is selected based on more than 3 decades of research. NGEN is a fully automatic (operator-free operation) that ensures peace of mind.

We deliver what we commit!!

Thank You

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Product Description

Product	Description	Make	Qty
nGen Air Package	nGEN 40 CP Nitrogen		1

Caption	Specification
Air Compressor KW	22 KW
Air Compressor Make	Atlas Copco/ Ingersoll Rand/ Gardner Denver/ As per Demand
Air Dryer	Included
Dry Air Storage	Included
Interconnecting Piping	Included
Pre filter	Included
Dry Air Generated	144 CFM

Product	Description	Make	Qty
nGen Nitrogen Generator	nGEN 40 Nitrogen		1

Caption	Specification
Gas Generated	Nitrogen
Gas Pressure	upto 6.5 Barg(Higher options Available)
Flow @ 95 %	107 Nm3/hr
Flow @ 97 %	98 Nm3/hr
Flow @ 98 %	87 Nm3/hr
Flow @ 99 %	80 Nm3/hr
Flow @ 99.5 %	71 Nm3/hr
Flow @ 99.9 %	64 Nm3/hr
Flow @ 99.95 %	54 Nm3/hr
Compressed Air Requirement	144 CFM
Module	PSA Plant
Air Compressor And Drier	In your Scope / Quoted Separately

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Contact: 1800 547 0506 | indsales@absstem.com
CIN: U29256HR2022PTC103387

Product	Description	Make	Qty
nGen Gas Storage	nGEN 40 ST Nitrogen		1

Caption	Specification
Water Volume	4000 Liters
Gas Stored	Nitrogen
Gas Stored Volume	26 M3
Working Pressure	6.5 Barg
Hydrotest Pressure	9 Barg

Product	Description	Make	Qty
nGen Air Package	nGEN 70 CP / nGen 65 CP Nitrogen		2

Caption	Specification
Air Compressor KW	37 KW
Air Compressor Make	Atlas Copco/ Ingersoll Rand/ Gardner Denver/ As per Demand
Air Dryer	Included
Dry Air Storage	Included
Interconnecting Piping	Included
Pre filter	Included
Dry Air Generated	250 CFM

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Product	Description	Make	Qty
nGen Nitrogen Generator	nGEN 70 Nitrogen		2

Caption	Specification
Gas Generated	Nitrogen
Gas Pressure	upto 6.5 Barg(Higher options Available)
Flow @ 95 %	185 Nm3/hr
Flow @ 97 %	168 Nm3/hr
Flow @ 98 %	150 Nm3/hr
Flow @ 99 %	138 Nm3/hr
Flow @ 99.5 %	125 Nm3/hr
Flow @ 99.9 %	110 Nm3/hr
Flow @ 99.95 %	93 Nm3/hr
Compressed Air Requirement	249 CFM
Module	PSA Plant
Air Compressor And Drier	In your Scope / Quoted Separately

Product	Description	Make	Qty
nGen Gas Storage	nGEN 70 ST Nitrogen	ABSSTEM	2

Caption	Specification
Water Volume	7000 Liters
Gas Stored	Nitrogen
Gas Stored Volume	38.5 M3
Working Pressure	5.5 Barg
Hydrotest Pressure	9 Barg

Commercial Offer

S No	Item Code	Item Name	Description	HSN	Qty	Unit Price	Discount	GST(%)	Amount
1	nGEN 40 CP	nGEN 40 CP	nGen Air Package	840510	1	10,11,000.00	2,78,025.00	18.00	7,32,975.00
2	nGEN 40	nGEN 40	nGen Nitrogen Generator	840510	1	20,80,800.00	5,72,220.00	18.00	15,08,580.00
3	nGEN 40 ST	nGEN 40 ST	nGen Gas Storage	840510	1	2,80,000.00	77,000.00	18.00	2,03,000.00

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4	nGEN 70 CP	nGEN 70 CP / nGen 65 CP	nGen Air Package	840510	2	15,68,000.00	4,31,200.00	18.00	22,73,600.00
5	nGEN 70	nGEN 70	nGen Nitrogen Generator	840510	2	29,90,000.00	8,22,250.00	18.00	43,35,500.00
6	nGEN 70 ST	nGEN 70 ST	nGen Gas Storage	840510	2	4,40,000.00	1,21,000.00	18.00	6,38,000.00

Sub Total	96,91,655.00
GST(%) @18.00	0.00
Grand Total ()	1,14,36,152.90

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Terms And Conditions

1.	EXCLUSIONS & CLIENTS SCOPE	All erection / civil work, All Power Cables & Cabling out of battery limit of PSA Plant, Cable Trenches etc. Unloading / Shifting of Gas Plant at Clients site will be in Client Scope. Additional gas piping, cooling water (if required).
2.	SUPERVISION OF ERECTION & COMMISSIONING	There shall be no commissioning charges, however you shall arrange freeTo & Fro Air Ticket, lodging / boarding, and local conveyance for our Engineers during supervision of Installation of Gas Plant at site on free of cost basis. Our Engineers will be there for Supervision only. Any manpower / fitter required for installation work shall be provided by you.
3.	P&F	3% Extra.
4.	Freight	Extra at Actuals.
5.	Insurance	In your Scope.
6.	Payment Terms	50% Advance with your Prestegious Purchase Order and Balance against PI.
7.	Delivery	6-8 Weeks from the PO and Advance. Rest can be confirmed for Availability.
8.	Warranty	We hereby confirm that, our equipment shall have Suppliers warranty for a period of 12 months from the date of Commissioning or 18 months from date of Dispatch whichever is earlier.
9.	GST	As per the Commercial Table.
10.	Remarks	.

Hope you will find our offer in line with your requirement.

Should you require any further clarification/information, please feel free to contact us.

Thanking you, and assuring you of our best attention at all times.

Very Yours truly,

For Absstem Industrial Solutions Pvt. Ltd.

Sushil Kumar

Senior Sales Engineer

9289722208

www.absstem.com

MINIMUM SPECIFICATIONS FOR FOOD GAS APPLICATIONS

Doc 126/18

Revision of Doc 126/11

EUROPEAN INDUSTRIAL GASES ASSOCIATION AISBL

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Doc 126/18

MINIMUM SPECIFICATIONS FOR FOOD GAS APPLICATIONS

Prepared by WG-8 Food gases and carbon dioxide

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Amendments from 126/11

Section	Change
	Editorial to align style with EIGA Style Manual
4.2	Expansion of section
6	Reference section updated
Appendix 1	Removal of reference to pharmaceutical specifications

Note: Technical changes from the previous edition are underlined

1 Introduction

Food gases are used as ingredients, additives and processing aids and include:

- carbonation of beverages);
- Modified atmosphere packaging;
- packaging (e.g. bread, meat etc.);
- storage (e.g. controlled atmospheres for fruit and vegetables); and
- processing (cooling, chilling, temperature controlling freezing, pH control etc.).

The minimum specifications for food gases in JECFA, European Union (EU) legislation are listed and this publication summarizes these standards (as applicable at the date of publication).

2 Scope

Specifications of all gases approved for use as food additives and, or used in other food applications.

3 Definitions

3.1 Assay

Purity of the gas.

3.2 Component / impurity

Specified characteristic of the product.

3.3 E xxx –

E - number for the product of the European Food Additive Regulations.

4 Application of food gases

The applications for gases in the food and beverage sectors fall into one of the three following categories:

4.1 Food additives

For gases to be used as a food additive, for example as a propellant or as a packaging gas, they shall be approved under EU law and are given E Numbers, for example E941 for nitrogen. The EU also sets minimum purity criteria for gases when used as a food additive. In addition to the purity criteria set down under EU Food Additive regulations, minimum specifications for gases are also published by JECFA (Joint FAO/WHO Expert Committee on Food Additives)

4.2 Food processing aids

Food Processing aids are legally defined as “Any substance not consumed as a food by itself, intentionally used in the processing of raw materials, foods or their ingredients to fulfil a certain technological purpose during treatment or processing, and that can result in the unintentional but technically unavoidable presence of residues of the substance or its derivatives in the final product, provided that these residues do not present any health risk and do not have any technological effect on the finished product”, see Regulation EC 1333/2008, *of the European Parliament and of the*

Council of 16 December 2008 on food additives [1][1]. Gases are processing aids when used during the processing of a food, for example liquid nitrogen or carbon dioxide for freezing, chilling and temperature control or inerting of bulk materials during processing but they are not themselves consumed as part of the food. In this case the only legal requirement is that the gas should not leave residues in the product that would present a risk to health.

NOTE Although food law requirements are generally applicable no purity criteria are set under EU law for use of gases as a processing aid. However national legislation can require a purity criteria alignment with those applied to food additives.

4.3 Food ingredients

A gas is described as an ingredient when it is used in the preparation of a food and is still present in the final product, even in an altered form, for example in the carbonation of beverages: Although food law requirements are generally applicable, no specific purity criteria are set under EU law for use of gases as an ingredient. Though any food additive criteria set for the gas could be relevant together with general food safety and hygiene legislation.

5 Specifications

See Appendix 1.

6 References

Unless otherwise specified the latest edition shall apply.

[1] Regulation EC 1333/2008, *of the European Parliament and of the Council of 16 December 2008 on food additives* www.europa.eu

[2] Commission Regulation (EU) No 231/2012 of 9 March 2012 laying down specifications for food additives listed in Annexes II and III to Regulation (EC) No 1333/2008 of the European Parliament and of the Council Text with EEA relevance

[3] JECFA – *Joint FAO/WHO Expert Committee on Food Additives, (FAO - Food and Agricultural Organisation of the United Nations and WHO - World Health Organisation);* www.fao.org

7 Other references

EC *Directive 2000/63/EC amending Directive 96/77/EC laying down specific purity criteria on food additives other than colours and sweeteners*; www.europa.eu

EC *Council Directive 98/83/EC of 3 November 1998 on the quality of water intended for human consumption;* www.europa.eu ,

EIGA Doc 70 *Carbon Dioxide Source Certification, Quality Standards and Verification;* www.eiga.eu

[1] References are shown by bracketed numbers and are listed in order of appearance in the reference section.

Appendix 1 - Summary of Current Gases Specifications in EU Legislation and JECFA

Component / Impurity	Standard	Carbon dioxide E 290	Nitrogen E 941	Oxygen E 948	Argon E 938	Helium E 939	Nitrous oxide E 942	n-/iso-Butane E 943 a/b	Propane E 944	Hydrogen E 949	Sulfur dioxide E 220
Assay (v/v)	EC	>99%	>99%	>99%	>99%	>99%	>99%	>96%/>94%	>95%	>99,9%	>99 %
	JECFA	>99%	>99%	>99%	>99%	>99%	>97%				>99,9 %
Odour	EC										
	JECFA			Free	free	free					
Moisture	EC		<0,05%	<0,05%	<0,05%	<0,05%	<0,05%	<50vppm	<50vppm	<50vppm	<0,05%
	JECFA	<52 vppm									<0,05%
CO_2	EC										
	JECFA			<300 vppm							
CO	EC	<10 vppm	<10 vppm				<30 vppm				
	JECFA	<10 vppm	<10 vppm	<10 vppm		<10 vppm	<10 vppm				
NO/NO_2	EC		<10 vppm				<10 vppm				
	JECFA						<5 vppm				
Total Hydrocarbon	EC		<100 vppm	<100 vppm	<100 vppm	<100 vppm		see *	see **		
	JECFA	<50vppm									
Residual Gases (O_2, N_2, H_2)	EC		< 1 % (O2)							<0.07%	
	JECFA				<1%						
Oil	EC	<5mg/kg									
	JECFA	<10 ppmw									
Acidity & Red. Subst.	EC	pass test									
	JECFA	pass test									
Halogens & H_2S	EC										
	JECFA						<5 vppm				
Other comp. & Heavy metals	EC										see***
	JECFA										see****

* Methane < 0,15%, other hydrocarbons < 5,1%. For E943a the maximum limit of hydrocarbons are:
C_2H_6 < 0,5%;
C_3H_8<1,5%;
i-C_4H_{10}<3%,
1.3-C_4H_6<0,1%

** Methane < 0,15%, other hydrocarbons are < 6,6%. For E943b the maximum limit of hydrocarbons are:
C_2H_6 < 0,5%;
C_3H_8<2,0%;
i-C_4H_{10}<4%,
1.3-C_4H_6<0,1%

*** EC Directive 2008/84 [2]: Non-volatile residue <0.001 %, Sulphur trioxide < 0,1 %, Selenium < 10 mg/kg, Arsenic <3mg/kg, Lead <5mg/kg, Mercury <1mg/kg, Heavy metals (as Pb) <10 mg/kg

**** JECFA (1998) [3]: Non-volatile residue <0,05%, Selenium <20 mg/kg, Lead <5mg/kg ,Other gases not present in air : No trace

nGEN

POWERED BY **ABSSTEM INDUSTRIAL**



THE NEXT-GENERATION OF INDUSTRIAL EXCELLENCE

Industrial on-site PSA Nitrogen Generator

Next-gen nitrogen PSA technology

With the **nGEN** Pressure Swing Adsorption (PSA) nitrogen generator, you ensure that your factory is using the latest clean technology, where the raw material is ambient air. The cost efficiency of a PSA nitrogen generator coupled with our easy-to-install with and-implement solution, make it **the preferred choice** over high-pressure cylinders and liquid nitrogen tanks.

Capitalise on the advantages of an on-site PSA nitrogen generator:

Non-stop availability for 24 hours a day, 7 days a week

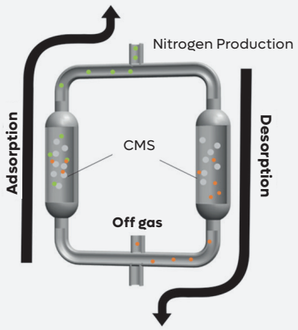
Easy integration with existing installations

Independent supply of industrial nitrogen

Reduced operational cost, no rental charges, logistical expenses and bulk user evaporation losses.

No safety hazards, hassle-free operation

The only raw material nGEN uses is atmospheric air



PSA (Pressure Swing Adsorption)
Process explaining Nitrogen Production

Yes, nGEN nitrogen generators only uses atmospheric air as the raw material.

The industrial nitrogen generator consists of two adsorbers filled with carbon molecular sieves (CMS). Pressurized air is passed through one of these adsorbers with Carbon, having a structure to attract oxygen, adsorbs oxygen from air and nitrogen is allowed to pass through to the storage tank and is produced as the final product. As the adsorber approaches saturation, the air is allowed to pass through the other adsorber while this adsorber is being regenerated to ensure a continuous supply of industrial nitrogen.

This changeover takes place through a specially designed micro controller to ensure a hassle-free operation. The quality of nitrogen produced is continuously monitored by a nitrogen analyzer.

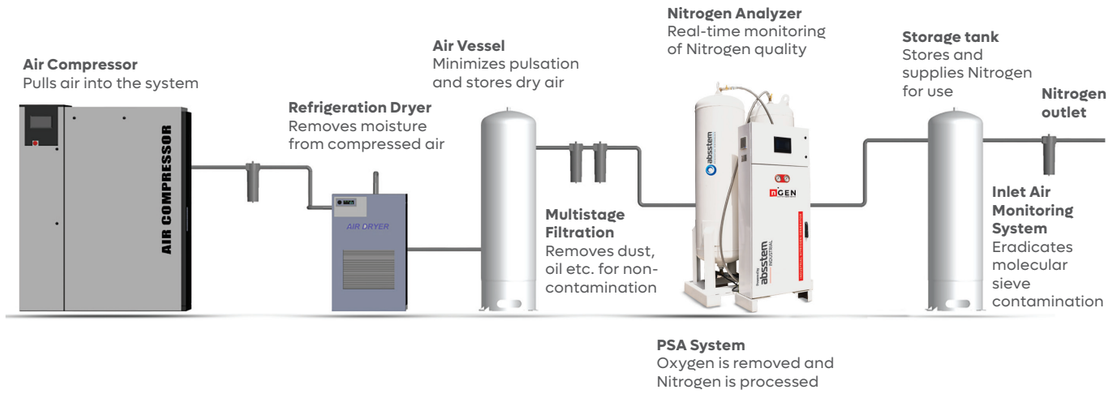
Certification and compliances

Our Quality Management Systems (QMS) is in full compliance with **ISO 9001:2015**

Our effective environmental management system (EMS) is in full compliance with **ISO 14001:2015**

Our occupational health and safety management systems are in full compliance with **ISO 45001:2018**

Redefining how industrial nitrogen is produced



Get the desired purity with nGEN nitrogen concentrations ranging from 99.5% to 99.9995%. Also achieve:

- Best-in-class ROI
- Lowered Operational Cost
- Quiet and silent operation
- Lowered carbon footprint
- Flexibility to upgrade to high purity onsite

Get the upper hand in operations using the following optional upgrades

- Nitrogen flow meter as standard
- Nitrogen sensor with a long lifetime
- Outlet pressure reducing valve
- Nitrogen pressure dewpoint sensor

nGEN is perfectly suited for a wide range of applications

- Food and Beverage Industry
- Laboratory Analysis
- Laser Cutting
- Electronics Industry
- Lighting Industry
- Chemical Blanketing
- Steel Manufacturing

Advanced features
Advanced capability

Reliable high flow capacity
The wide product range and gas flows exceeding 3,000 Nm³/h make nGEN ideal for a variety of demanding applications.

Exceptional performance in tough weather conditions
- 24x7 Continuous production
- Built to last extreme weather conditions and tough terrains
- Show longevity in a dust/moisture-laden environment
- Ergonomic design, robust structure and advanced HMI
- Production Breakdown risks and downtimes are nullified

Proven use-cases, it's always Plug and Play
- Fully automated and monitored including the Nitrogen sensor
- The only inputs required are air and electricity
- Get the desired purity your application requires

Achieve best-in-Class ROI
- Very Low operating expenses
- No additional costs such as order processing, refills and delivery charges
- Limited maintenance costs



Technical Specifications

Relative humidity	Max 95%
Nitrogen Flow	2.5 to 3000 - NM³/hr
Working temperature	- 10°C to 55°C
Power Supply	230 VAC, 50/60 Hz
Inlet air pressure	Min 7.0 barg*
N_2 Delivery Pressure	Up to 5.5 barg#
Nitrogen Dew Point	- 50°C
ISO 9001:2015	Manufacturing is in accordance with Certified Quality Management Systems

** Higher pressure available on demand*
Higher dew point available on demand

Trusted and proven industry-leading technical features

Self-protective monitoring of the feed air quality parameters
- Temperature
- Pressure
- Pressure dewpoint
- Automatic feed air blow-off in case of contamination

Advanced HD Touch Screen Display
- Fully automatic, user friendly
- Remote monitoring equipped
- Advanced GUI with data monitoring and storage
- Nitrogen purity readings in real-time
- Triggered alerts and alarms on any abnormal functionality
- Data storage with external storage capability for audit purposes
  - Can be remotely activated
  - RS232, RS422, USB and Ethernet ports supported

Highest Quality Carbon Molecular Sieve (CMS)
- Very High Density due to packed-bed Technology
- Protected by the dedicated pressure sensor
- 10 year warranty on CMS*

Multistage filtration system
- Filters remove oil and dust particles
- Additional filters drain moisture using an ADV (Automatic Drain Valve)

Automatic start-up
- Eliminates the risk of overflow and CMS damage

Premium energy efficiency
- Lowest Air factor (Air-to-Nitrogen ratio) from 1.8 (95% N_2) in the industry

The ultimate energy saver
- Self-regulation and stable purity
- Stand-by mode in case no Nitrogen is consumed
- Automatically regulates the requested nitrogen pressure and purity

** When the plant is under CMC*

LOWEST SPECIFIC POWER in the industry

Inlet air as per **ISO 8573.1:2010 - CLASS 1.4.1**

PURITY TESTING DONE by NABL APPROVED lab

Pressure up to **350 KG/CM²G** is available on-demand through Nitrogen compressors

SOUND LEVEL OF 65 DB from a distance of 1 meter

USB STORAGE to store parameters values

CE-CERTIFIED ELECTRONICS with 24 VDC Operating Voltage

All tanks are fabricated per **ASME STANDARDS**

CORROSION-RESISTANT internal environment for all tanks

No more of the old industrial challenges

Avoid high-pressure cylinders and liquid nitrogen tanks.

Put an end to the 5-major issues plaguing your industry such as:

Unprecedented Price Hikes
Bottled and liquified gases are subject to fluctuating prices.

Demand & Supply Gap
The highly dynamic nature of the market, with multiple vendors having different supply capabilities, and distribution at business-critical moments, can cause a large gap in supply when demand rises.

Remote Location Access
The more remote the location, the greater the transportation and logistical costs.

Evaporation Losses
5-10% of evaporation and non-usage losses are observed.

Storage & Handling Safety Concerns
Nitrogen cylinders have safety hazards and need to be handled by a well-trained team.

nGEN solves these issues cost-efficiently and at business speed, delivering consistent performance while assuring absolute reliability for your industrial requirement.

nGEN offers diverse Packaged Solutions

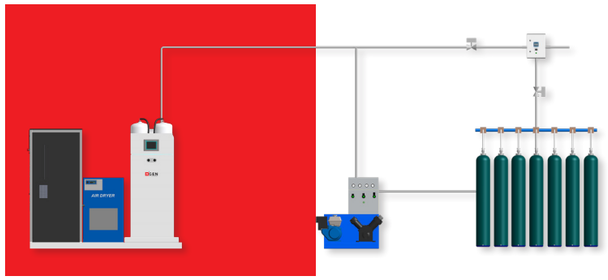


PSA Nitrogen Generator
This is a plug-and-play system that is connected to your air compressor and storage tanks.



Skid Package System
This system comes with an air compressor, multistage filtration system and a de-greased storage vessel. This is a complete package with all instruments required for nitrogen generation.

High-pressure Nitrogen Cylinder Filling System
nGEN with a Nitrogen Booster for filling of cylinders.
Pressure – Up to 150 barg
Filling range 8-1000 cylinders/day





Containerized Mobile PSA Nitrogen Generator
Skid-mounted mobile systems with complete piping and interlocking support. They can be transported to any location and designed according to climate conditions and environmental factors.

Note:- Cylinder filling options are available with noise control.

Choose the model that best fits your requirements.

Models	Purity (NM³/hr)						
	95%	97.00%	98.00%	99.00%	99.50%	99.90%	99.95%
nGEN 01	8	7	6	5.5	5	4	3
nGEN 02	10	9	8	7.5	6	5	4
nGEN 04	14	12	11	10	8	7	6
nGEN 05	21	18	16	15	13	11	8
nGEN 07	28	24	22	20	17	14	11
nGEN 08	35	30	27	25	21	18	14
nGEN 09	38	33	30	27	23	20	15
nGEN 10	46	40	36	33	28	24	18
nGEN 20	66	57	52	48	40	34	26
nGEN 30	75	65	59	54	45	39	30
nGEN 40	90	78	71	65	55	47	36
nGEN 50	111	96	88	80	67	58	44
nGEN 60	135	117	107	98	82	70	54
nGEN 70	165	143	130	119	100	86	66
nGEN 80	205	177	161	148	124	106	82
nGEN 90	234	203	184	169	141	122	94
nGEN 100	338	293	266	244	204	176	135
nGEN 110	391	339	308	283	237	203	157
nGEN 120	485	420	382	350	293	252	194
nGEN 130	579	501	456	418	350	301	231
nGEN 140	676	586	533	488	409	352	270
nGEN 150	787	682	620	568	476	409	315
nGEN 160	1047	907	825	756	633	544	419
nGEN 170	1242	1077	979	897	751	646	497

Models	Purity (NM³/hr)			
	99.99%	99.995%	99.999%	99.9995%
nGEN 01 P	2.4	1.5	1.2	1
nGEN 05 P	7	3.9	3.2	2.6
nGEN 10 P	14	9	7	6
nGEN 20 P	21	12	10	8
nGEN 30 P	23	14	11	9
nGEN 40 P	28	17	14	11
nGEN 50 P	35	21	17	14
nGEN 60 P	42	25	21	17
nGEN 70 P	52	31	25	20
nGEN 80 P	64	38	31	25
nGEN 90 P	73	43	36	29
nGEN 100 P	105	63	52	42
nGEN 110 P	122	73	60	48
nGEN 120 P	151	90	74	60
nGEN 130 P	181	107	88	72
nGEN 140 P	211	126	103	84
nGEN 150 P	246	146	120	97
nGEN 160 P	327	194	160	130
nGEN 170 P	388	231	190	154

A world-standard plan, with pan-India service.

Absstem Shield: Premium Service across India

Absstem is a truly national organisation with support available in more than 25 states. As a result, our field service engineers are never far away. We pride ourselves on the swift assistance that keeps your nGEN system performing reliably and efficiently.

Our Service Plans keep your nGEN nitrogen systems in excellent shape. We have options that could look after your installations for as long as 10 years

10 year CMC Plans

We have CMC solutions that look after your machines for as long as 10 years

Available 24/7

We keep spare parts in stock so your machine is up and running as quickly as possible

Quickest Turn Around Time in the industry

Our shield engineer is assigned to you within 10 minutes

For specific upgrades, add-ons and customised requirements, please speak to us.

Robust countrywide presence

200+ robust footprint nationwide.

We have covered over 31 states & union territories in India.

Our wide service network covers expand to all states.

absstem
REINVENTING PERFORMANCE

About Absstem

Innovation is as essential to our team as nitrogen is to your business. So how do we constantly innovate? By disrupting the existing category codes. We invest not only in gas separation of the purest grade but also maintain high standards in our after sales service. Pioneering the PSA technology in India, we deal with complete range of gas separation plants. Keeping the pioneer spirit alive and well, we have become the market leaders in gas separation.

When it comes to the industrial sector, the requirements are always specific and the applications always demand a specific focus. The solutions to the problems are met with improved outputs and better efficiency - making it a success story for all parties involved. Absstem's research and development team took this into account and conceptualized a sustainable engineering solution for the nitrogen requirement which took care of their problem statements such as operator needs, consistent flow and continuous purity levels.

Best known for our presence in the on-site oxygen generation in the Indian industrial sector, we are also recognised by Best known for our presence in the on-site oxygen generation in Indian industrial sector we are also recognised by Dept. of Industrial Policy & Promotion (DIPP) for innovation in the industrial sector and registered under the Ministry of Micro, Small and Medium Enterprises, Government of India (MSME) and Confederation of Indian Industry (CII).

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MAKE IN INDIA

MSME
सूक्ष्म, लघु एवं मध्यम उद्यम
MICRO, SMALL & MEDIUM ENTERPRISES

CII
Confederation of Indian Industry

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